Satisfiability Of A Single Member Equality

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1 Setup

The ground alphabet of members is some set of literals L, while the non-ground variables are V. A substitution σ is a function $V \to (L \cup V)^*$ that acts on strings by substituting each variable in place, so $\sigma(e_1 v e_2) = \sigma(e_1)\sigma(v)\sigma(e_2)$. For a ground substitution the image is restricted to L^*. A (ground) substitution σ is called proper for s if $\sigma(s)$ contains no duplicate literals. A basic example of a ground substitution is $\ell : (L \cup V)^* \to L^*$ defined by sending all variables to the empty string ε.

A model for the equality problem of two strings s_1, s_2 in $(L \cup V)^*$ is a proper ground substitution (for s_1 and s_2) where the strings become equal $\sigma(s_1) = \sigma(s_2)$. Since no proper substitution exists when a string already contains a duplicate literal we restrict ourselves to strings without duplicate literals.

The main result is Theorem 1 giving exact conditions for the existence of a model. Section 3 has no application currently and is kept only for future reference.

2 Analysis

Lemma 2.1 (Minimal alphabet)**.** *Let L_{min} be the set of all the distinct literals appearing in s_1 and s_2; then there exists a model iff there exists a model over L_{min}.*

Proof. For a satisfying model σ define $\sigma_{\min}$ by deleting all literals not in $L_{\min}$ from the image. Then $\sigma_{\min}(s_1) = \sigma_{\min}(s_2)$ is obtained from $\sigma(s_1) = \sigma(s_2)$ by likewise deleting all literals not in $L_{\min}$ from both sides, which does not change the result. □

Lemma 2.2 (Duplicate variables)**.** *If there exists a model then there also exists a model where all variables that occur more than once in total across s_1 and s_2 are sent to the empty string.*

Proof. If a variable occurs more than once in the same string any proper ground substitution must send it to the empty string. For the remaining case of a variable v appearing exactly once in each string, properness for s_1 and s_2 dictates that the literals of $\sigma(v)$ are distinct from all literals appearing in s_1 or s_2. By Lemma 2.1 such literals can be discarded. □

Lemma 2.3 (Duplicate-free case)**.** *If there are no duplicate literals and variables across s_1 and s_2 then there exists a model exactly unless:*

1. *s_1 and s_2 both start with a literal*

2. s_1 and s_2 both end with a literal

3. s_1 contains no variables and s_2 contains a literal

4. s_2 contains no variables and s_1 contains a literal

Proof. It is easy to see that if any of the four rules is true then no model can exist. It remains to show that a model exists if all conditions are false.

First distinguish on whether s_1 and s_2 are empty. If $s_1 = s_2 = \varepsilon$ then all four rules are false and a model exists. When only one string is empty a model exists iff the other string contains no literals. This is exactly asserted by rules (3) and (4) while rules (1) and (2) are automatically false.

For s_1 and s_2 both non-empty rules (1) and (2)imply that at least one of the two strings starts with a variable and at least one of them ends with a variable. If each string has at least one variable at its start or end then the problem must be of the form $\sigma(v_1e_2) = \sigma(e_1v_2)$ which is solved by $\sigma(v_1) = \ell(e_1)$, $\sigma(v_2) = \ell(e_2)$ and $\sigma(v) = \varepsilon$ for $v \neq v_1, v_2$.

The remaining case is that one string (say s_1) has literals at its start and end position, so the other string (say s_2) must have variables at both. If s_2 contains a literal then by rule (3) s_1 must contain a variable. This gives the problem shape $\sigma(e_1v_2e_3) = \sigma(v_1e_2v_3)$ with solution $\sigma(v_i) = \ell(e_i)$ and $\sigma(v) = \varepsilon$ otherwise. When s_2 contains no literals then let v_0 be one of its variables and define $\sigma(v_0) = \ell(s_1)$ and $\sigma(v) = \varepsilon$ for $v \neq v_0$, which solves $\sigma(s_1) = \ell(s_1) = \sigma(s_2)$. □

Theorem 1. *There exists a model for s_1 and s_2 iff all literals shared between s_1 and s_2 appear in the same order in both strings and the matching (duplicate-free) parts before, between and after the shared literals meet the criterion of Lemma 2.3 after deleting all duplicate variables.*

Proof. No model can introduce a literal that already appears in s_1 or s_2 without introducing duplicates. So the only thing that can match a shared literal in s_1 is the same literal in s_2. And they can only be matched in order, justifying the first requirement.

If all shared literals match in order the parts before, after and between them must also match. These parts do not contain duplicate literals by construction and duplicate variables can be removed without affecting the existence of a model by Lemma 2.2. Lemma 2.3 gives the exact conditions for model existence of such strings. □

3 Ordering

Define the following two string predicates:

$$E(s_1, s_2) \coloneqq \exists \sigma \text{ proper ground substitution} : \sigma(s_1) = \sigma(s_2)$$
$$E^*(s_1, s_2) \coloneqq \exists \sigma_1, \sigma_2 \text{ proper ground substitutions} : \sigma_1(s_1) = \sigma_2(s_2)$$

E describes the existence of a model for the equality problem of s_1, s_2. E^* describes the existence of such a model when shared variables between s_1 and s_2 are not taken into account. Obviously $E(s_1, s_2) \implies E^*(s_1, s_2)$.

Next consider the ordering predicates:

$$R(s_1, s_2) \coloneqq \forall q \in (L \cup V)^* : E(s_1, q) \Rightarrow E(s_2, q)$$
$$R^*(s_1, s_2) \coloneqq \forall q \in (L \cup V)^* : E^*(s_1, q) \Rightarrow E^*(s_2, q)$$

Lemma 3.1 (Partial order)**.** *R and R^* are partial orderings on strings modulo the equivalence relations $\sim$ and $\sim^*$ respectively, where*

$$s_1 \sim s_2 \coloneqq \forall q : E(s_1, q) \Leftrightarrow E(s_2, q)$$
$$s_1 \sim^* s_2 \coloneqq \forall q : E^*(s_1, q) \Leftrightarrow E^*(s_2, q)$$

Proof. The properties $R(s, s)$, $R(s_1, s_2) \wedge R(s_2, s_1) \iff s_1 \sim s_2$ and $R(s_1, s_2) \wedge R(s_2, s_3) \implies R(s_1, s_3)$ fall out of the logical rules for implications. The same is true for R^*. □

Lemma 3.2. $R^*(s_1, s_2) \not\Longrightarrow R(s_1, s_2)$

Proof. Consider $s_1 = l$ and $s_2 = vl$. $R^*(s_1, s_2)$ is true because from $\sigma_2(q) = \sigma_1(s_1) = l$ we have that $\sigma_1'(v) = \ell(v)$ satisfies $\sigma_1'(s_2) = \sigma_1'(vl) = l = \sigma_2(q)$. However $R(s_1, s_2)$ is false because for $q = v$, $\sigma(s_1) = l = \sigma(v)$ is satisfiable but $\sigma(s_2) = \sigma(vl) = \sigma(v) = \sigma(q)$ is not. □

Lemma 3.3 (Ground tests)**.** *Let $\mathcal{I}(s) \coloneqq \{\sigma(s) \mid \sigma$ proper ground substitution$\} \subseteq L^*$ be the (duplicate-free) instances of s. Then $w \in \mathcal{I}(s) \iff E(s, w) \iff E^*(s, w)$.*

Proof. w contains no variables, so $\sigma(w) = w$ for every σ. Hence whether one substitution or two independent ones are used is immaterial: both $E(s, w)$ and $E^*(s, w)$ state exactly that some model σ satisfies $\sigma(s) = w$. □

Lemma 3.4. *If $R(s_1, s_2)$ or $R^*(s_1, s_2)$ then $\mathcal{I}(s_1) \subseteq \mathcal{I}(s_2)$.*

Proof. By using $R(s_1, s_2)$ or $R^*(s_1, s_2)$ with $q = w \in \mathcal{I}(s_1)$ we have

$$w \in \mathcal{I}(s_1) \implies E^{(*)}(s_1, w) \implies E^{(*)}(s_2, w) \implies w \in \mathcal{I}(s_2).$$

□

Lemma 3.5 (Characterisation of R^*)**.** $R^*(s_1, s_2) \iff \mathcal{I}(s_1) \subseteq \mathcal{I}(s_2)$.

Proof. ($\Rightarrow$) By Lemma 3.4. ($\Leftarrow$) Let q be arbitrary with $E^*(s_1, q)$, witnessed by σ_1, σ_2 with $w \coloneqq \sigma_1(s_1) = \sigma_2(q)$. From $w \in \mathcal{I}(s_1) \subseteq \mathcal{I}(s_2)$ it follows that there exists σ_1' with $\sigma_1'(s_2) = w = \sigma_2(q)$, which is $E^*(s_2, q)$. □

Lemma 3.6 (R^* extends R)**.** $R(s_1, s_2) \implies R^*(s_1, s_2)$.

Proof. $R(s_1, s_2) \implies \mathcal{I}(s_1) \subseteq \mathcal{I}(s_2) \implies R^*(s_1, s_2)$ □

Lemma 3.7 (Substitutions)**.** *If $\rho : V \to (L \cup V)^*$ is a proper substitution for s then $R^*(\rho(s), s)$.*

Proof. Let q be arbitrary with $E^*(\rho(s), q)$ witnessed by σ_1 and σ_2. Then $\sigma_1'(v) = \sigma_1(\rho(v))$ satisfies $\sigma_1'(s) = \sigma_1(\rho(s)) = \sigma_2(q)$, which is $E^*(s, q)$. □

Lemma 3.8 (Min property of ℓ)**.** *If s contains no duplicate literals then* $\min_{w\in\mathcal{I}(s)} |w| = |\ell(s)|$.

Proof. For any proper ground substitution σ, deleting all literals from its images can only make $\sigma(s)$ shorter, so $|\sigma(s)| \geq |\ell(s)|$ and $\min_{w\in\mathcal{I}(s)} |w| = \min_\sigma |\sigma(s)| \geq |\ell(s)|$. But at the same time $\min_{w\in\mathcal{I}(s)} |w| \leq |\ell(s)|$ because $\ell(s) \in \mathcal{I}(s)$, as ℓ is itself a proper ground substitution for s. □

Lemma 3.9 (R^ℓ extends R^*)**.** *Let* $R^\ell(s_1, s_2) \coloneqq |\ell(s_1)| \geq |\ell(s_2)|$*, then* $R^*(s_1, s_2) \implies R^\ell(s_1, s_2)$.

Proof. From $\ell(s_1) \in \mathcal{I}(s_1)$ and $\mathcal{I}(s_1) \subseteq \mathcal{I}(s_2)$ it follows that $\ell(s_1) \in \mathcal{I}(s_2)$ and that $|\ell(s_1)| \geq \min_{w \in \mathcal{I}(s_2)} |w| = |\ell(s_2)|$. $\square$